UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

PHYSICS **5054/01**

Paper 1 Multiple Choice **October/November 2008**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

9293832192

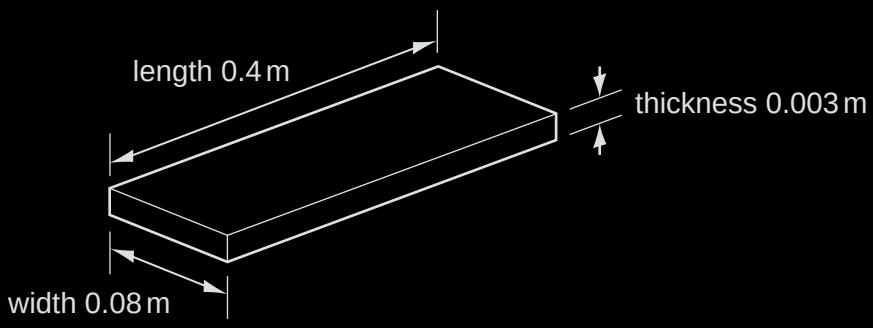
READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

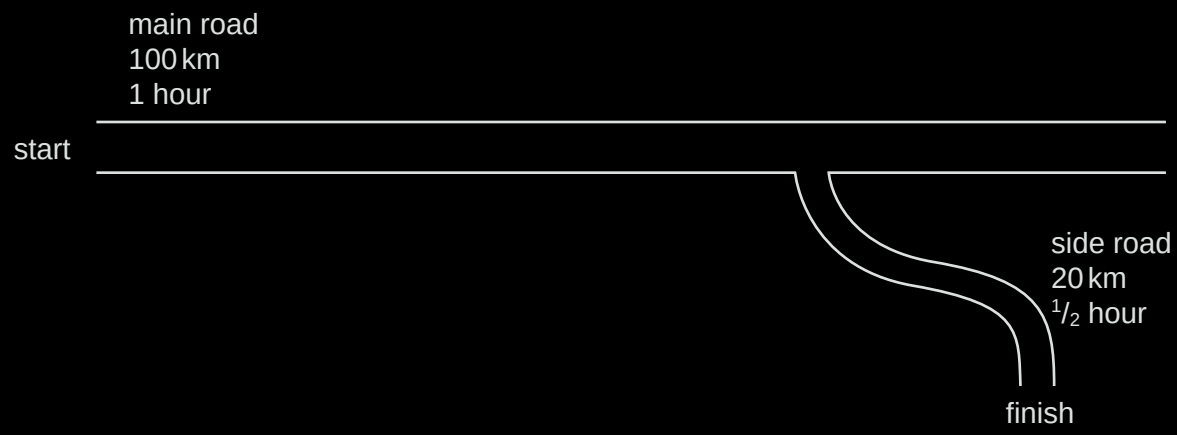
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.



Which instruments measure each of these dimensions accurately?

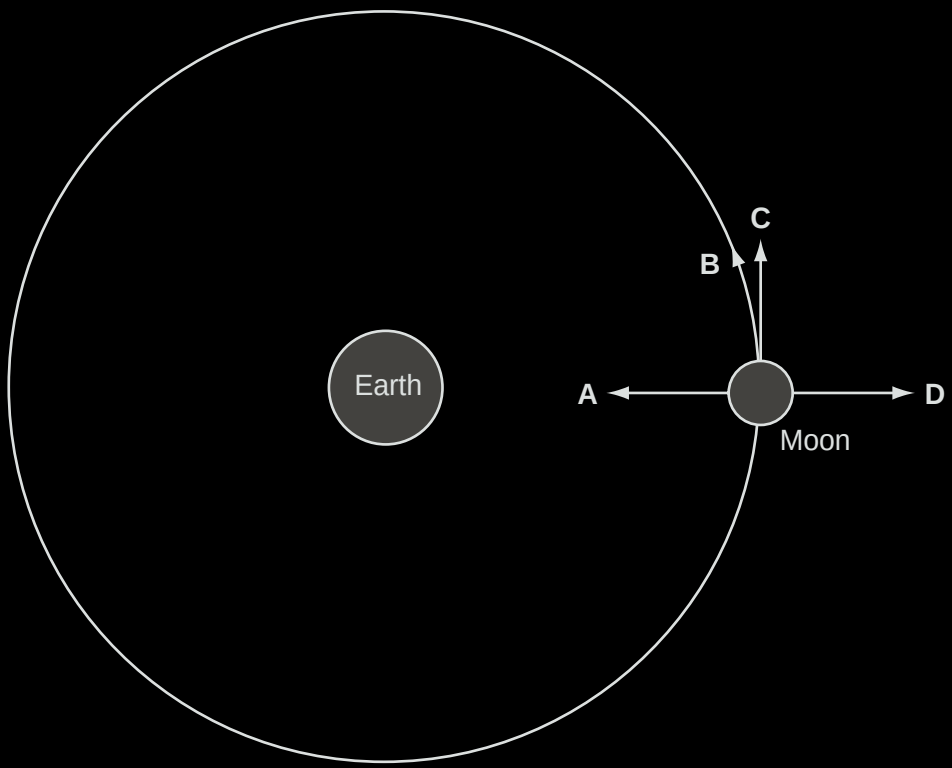
	length	thickness	width
A	metre rule	micrometer	vernier calipers
B	metre rule	vernier calipers	micrometer
C	micrometer	metre rule	vernier calipers
D	vernier calipers	micrometer	metre rule

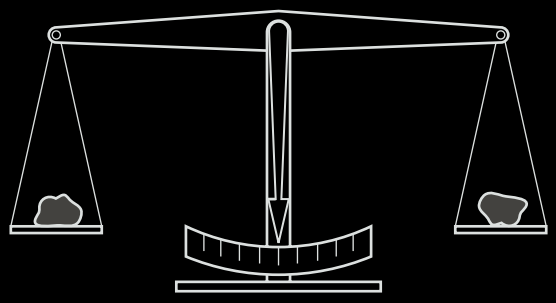
2 A car takes 1 hour to travel 100 km along a main road and then ½ hour to travel 20 km along a side road.



What is the average speed of the car for the whole journey?

A 60 km / h **B** 70 km / h **C** 80 km / h **D** 100 km / h


C
B
Earth
A
D
Moon



Which need not be the same?

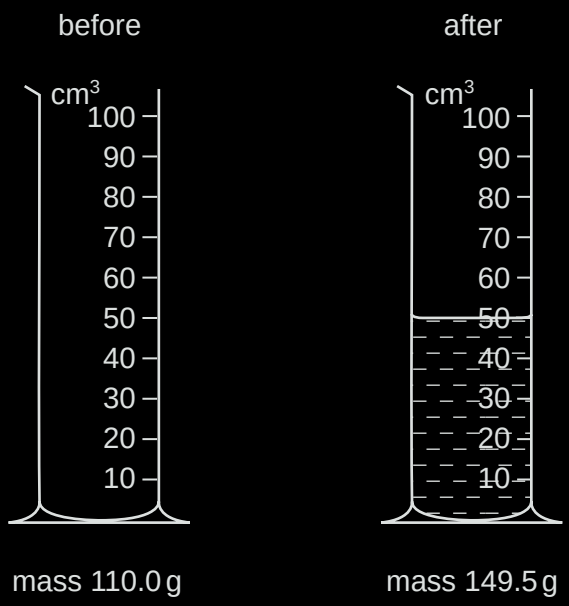
A the masses of the two objects

B the moments about the pivot of the two objects

C the volumes of the two objects

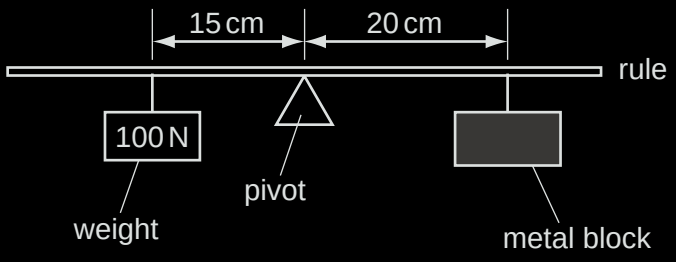
D the weights of the two objects

6 The mass of a measuring cylinder is measured before and after pouring a liquid into it.



What is the density of the liquid?

A 0.79 g / cm^3 **B** 1.3 g / cm^3 **C** 1.4 g / cm^3 **D** 2.2 g / cm^3

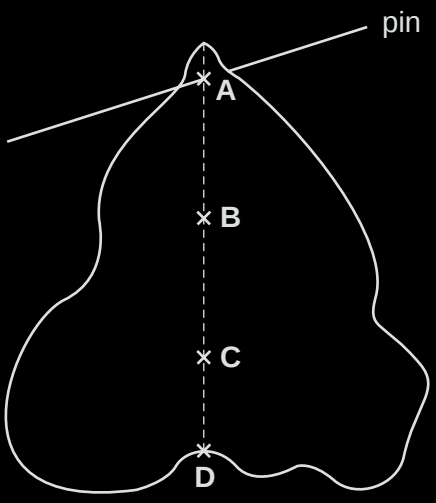


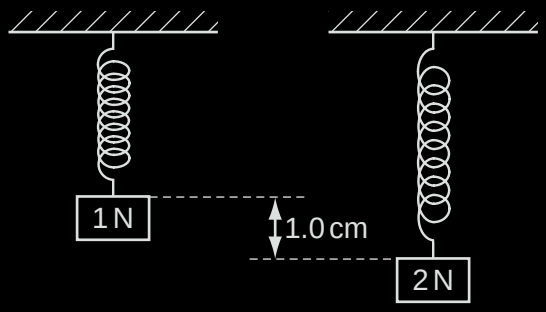
What is the weight of the metal block?

A 50 N **B** 75 N **C** 100 N **D** 150 N

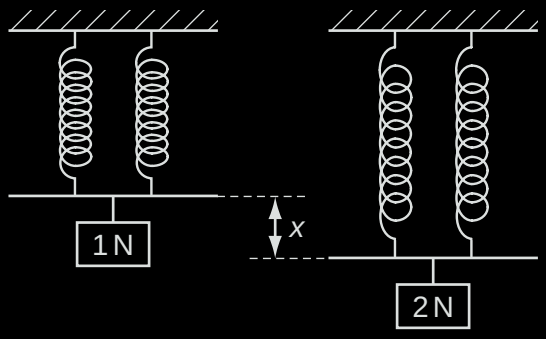
8 A piece of uniform card is suspended freely from a horizontal pin.

At which of the points shown is its centre of gravity?





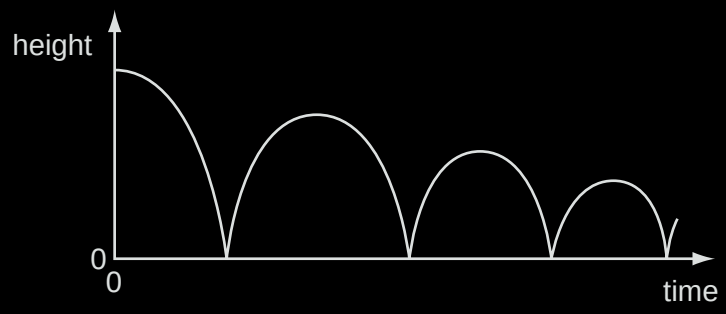
Two springs that are identical to the first one are put side by side. They are connected at both ends, and a 1 N weight is hung on them. The load is then increased to 2 N.



What is the new increase *x* in the extension?

A 0.5 cm **B** 1.0 cm **C** 2.0 cm **D** 3.0 cm

10 The graph shows how the height above the ground of a bouncing ball changes with time.



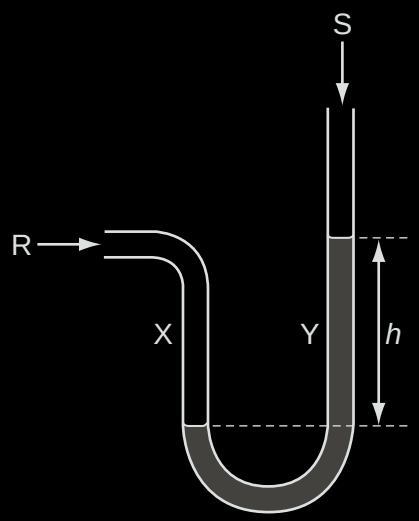
Which statement explains why the height of each peak decreases with time?

A Kinetic energy is converted to potential energy at each bounce.

B Kinetic energy is converted to thermal energy at each bounce.

C The ball gains energy on impact with the floor.

D The ball is wearing out.



Side X is connected to a gas supply. Side Y is open to the atmosphere.

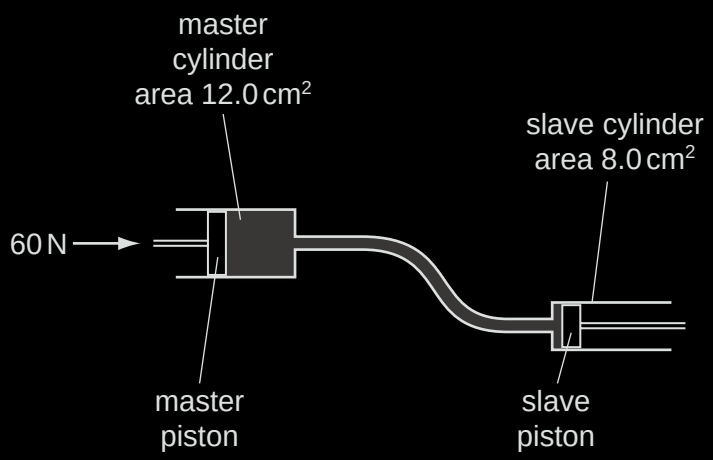
What pressure is the length *h* used to measure?

A the atmospheric pressure S

B the difference between the gas pressure R and the atmospheric pressure S

C the gas pressure R

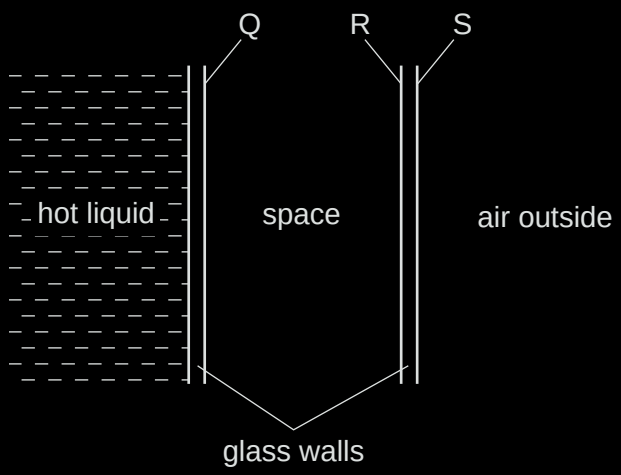
D the sum of the gas pressure R and the atmospheric pressure S



The cross-sectional area of the master cylinder is 12.0 cm^2 and the cross-sectional area of the slave cylinder is 8.0 cm^2. The force applied to the master piston is 60 N, producing a pressure of 5.0 N/cm^2.

Which line in the table is correct?

	pressure at slave cylinder N/cm^2	force at slave cylinder N
A	3.3	40
B	3.3	90
C	5.0	40
D	5.0	90



What reduces heat losses by radiation?

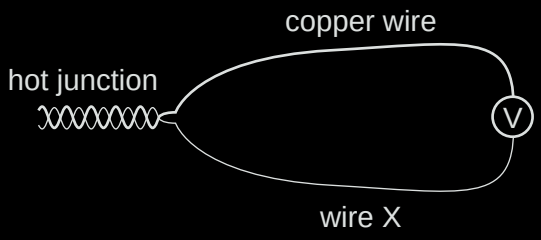
A a vacuum in the space between the walls

B painting surface Q black

C painting surface R black

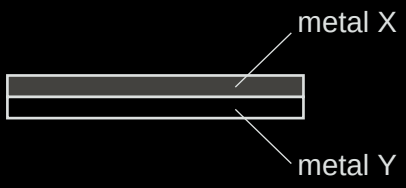
D painting surface S silver

17 A thermocouple thermometer is made from two wires connected to a voltmeter.

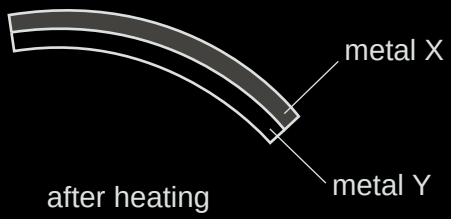


Which arrangement gives a reading on the voltmeter?

	temperature of voltmeter	wire X
A	colder than hot junction	copper
B	colder than hot junction	iron
C	same as hot junction	copper
D	same as hot junction	iron



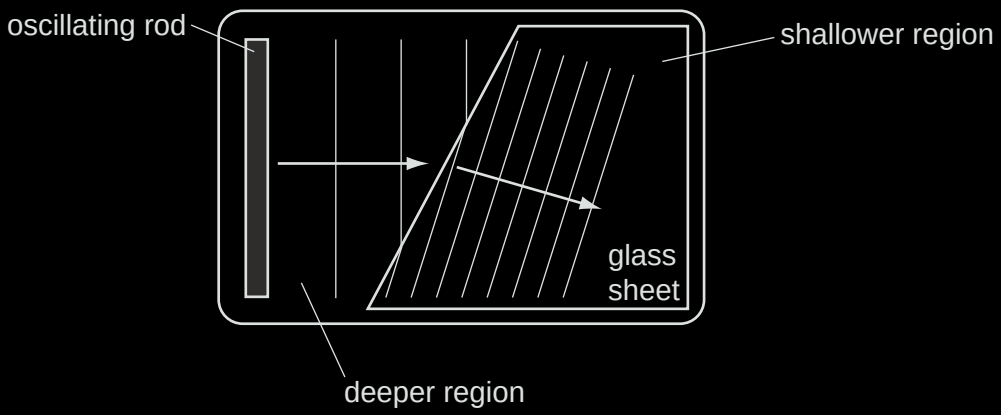
at room temperature



after heating

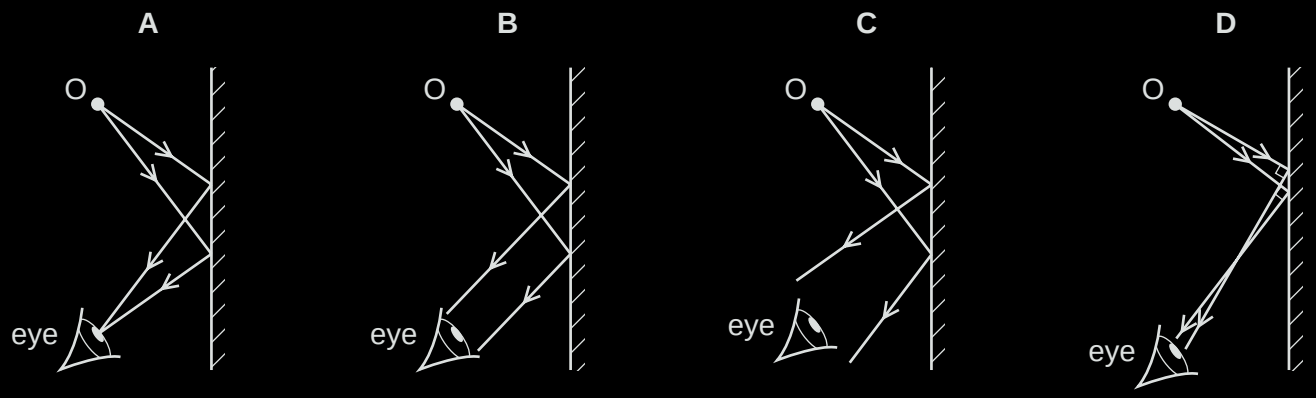
Which metals produce the above effect?

	metal X	metal Y
A	aluminium	platinum
B	copper	aluminium
C	steel	copper
D	platinum	steel

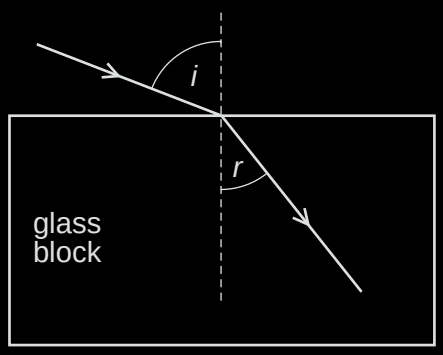


When crossing into the shallower region, what is the effect on the frequency and on the speed of the waves?

	wave frequency	wave speed
A	changes	changes
B	changes	unchanged
C	unchanged	changes
D	unchanged	unchanged

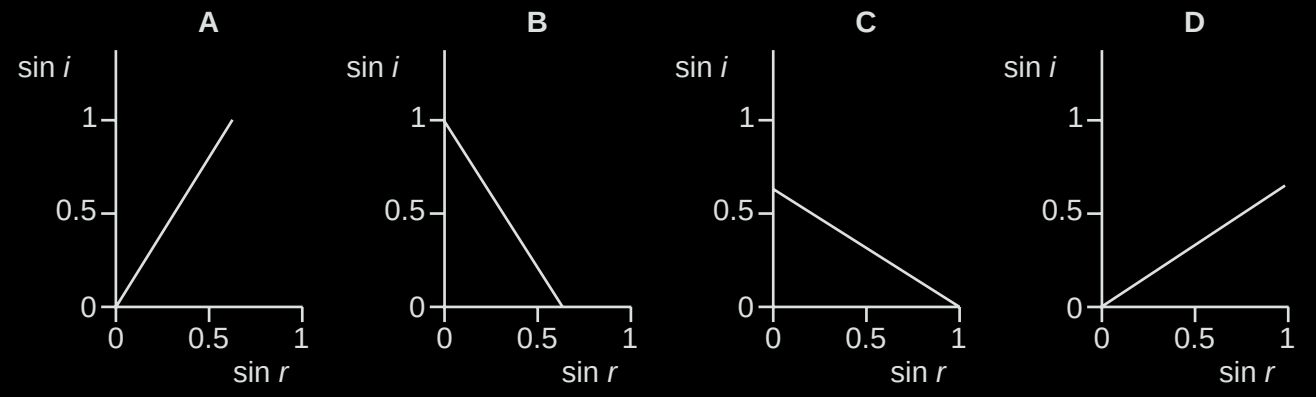


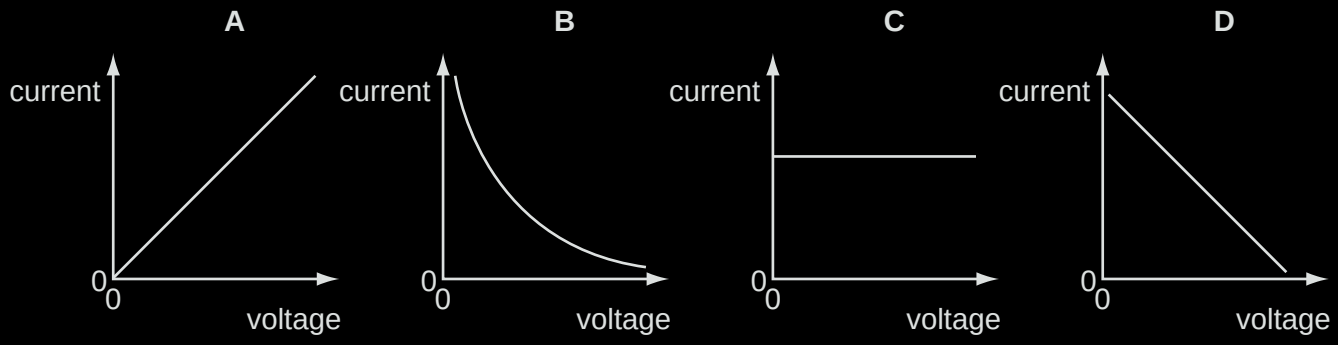
24 A ray of light enters a glass block at an angle of incidence *i* producing an angle of refraction *r* in the glass.



Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

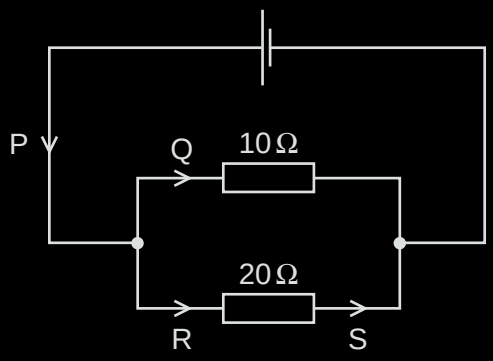
Which graph is correct?





30 The circuit diagram shows a parallel arrangement of resistors.

P, Q, R and S represent the current at the points shown.



Which statement is correct?

A P is greater than Q.

B Q is equal to R.

C R is greater than S.

D S is equal to P.

31 An electric heater is rated at 3 kW. Electrical energy costs 20 cents per kW h.

What is the cost of using the heater for five hours?

A 12 cents **B** 60 cents **C** 100 cents **D** 300 cents

32 What is the purpose of a circuit breaker in an electric circuit?

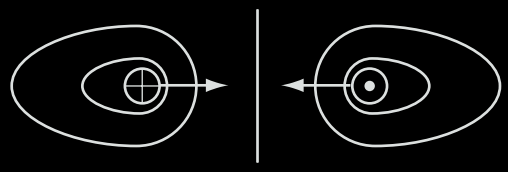
A to change alternating current into direct current

B to keep the current constant

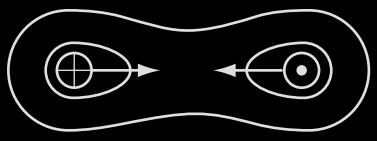
C to prevent the current from becoming too large

D to reduce the current to a safe value

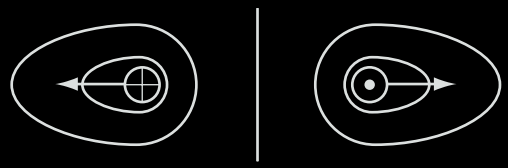
A



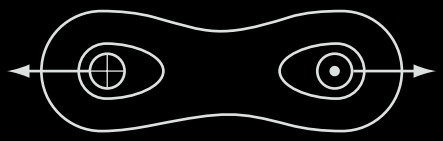
B



C



D



key

⊕ conductor carrying current into page

⊙ conductor carrying current out of page

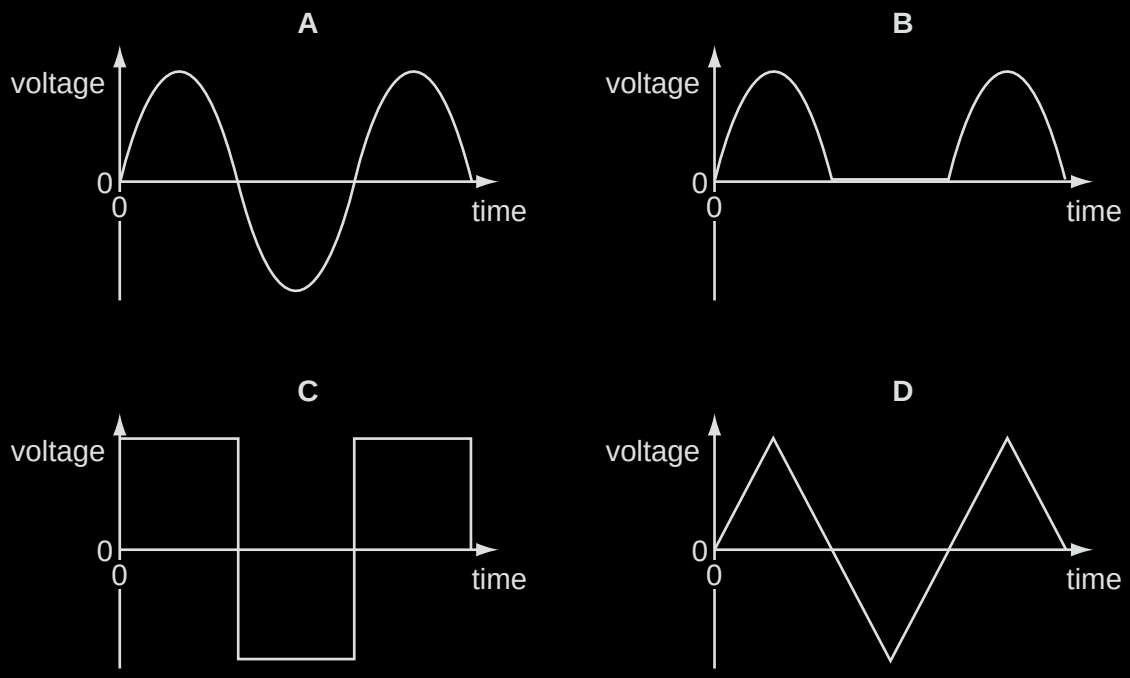
34 The e.m.f. induced in a conductor moving at right-angles to a magnetic field does **not** depend upon

A the length of the conductor.

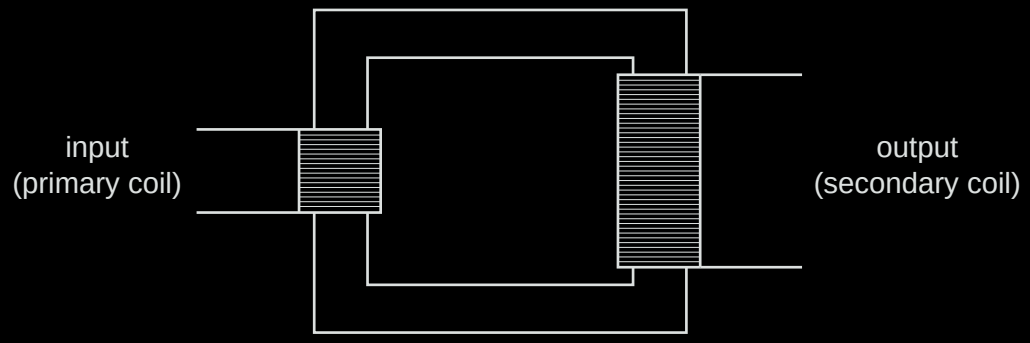
B the resistance of the conductor.

C the speed of the conductor.

D the strength of the magnetic field.



36 The diagram shows a working transformer.



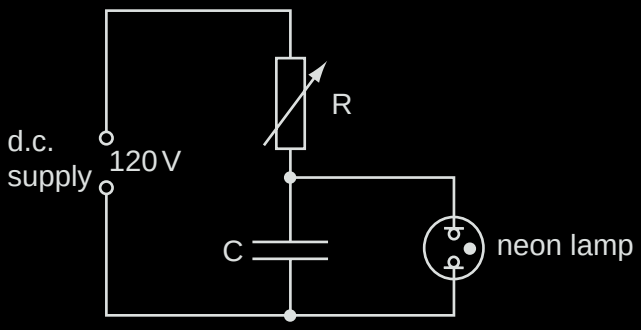
Which statement is correct?

A The input voltage is d.c.

B The input voltage is greater than the output voltage.

C The input voltage is less than the output voltage.

D The input voltage is the same as the output voltage.



The neon lamp lights when the p.d. across the capacitor C reaches 110 V. C then discharges through the neon lamp. When C has discharged, the lamp goes out and the p.d. across C slowly increases once more.

What is the effect of increasing the resistance of R?

A The lamp lights up and goes out at the same rate.

B The lamp lights up and goes out less frequently.

C The lamp lights up and goes out more frequently.

D The lamp will not go out at all.

38 What occurs in the decay of a radioactive nucleus?

A The nucleus absorbs another nucleus.

B The nucleus absorbs at least one form of radiation.

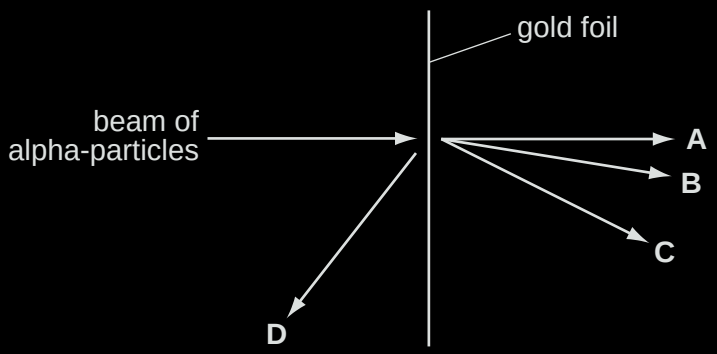
C The nucleus always splits into two equal fragments.

D The nucleus emits at least one form of radiation.

39 The radioactive nucleus $^{87}_{37}\mathrm{Rb}$ decays with the emission of a beta-particle.

How many protons and neutrons are left in the nucleus?

	protons	neutrons
A	37	49
B	37	50
C	38	49
D	38	87


gold foil
beam of
alpha-particles
A
B
C
D